

Wrocław University of Science and Technology
Faculty of Management

Descriptive analytics

Fourth task set report
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1 Task 4.1

1.1 Description

The task included simulating an exponential smooth transition regression model and the subsequent evaluation of its least squares loss score. The underlying data generating process relied on a transition variable defined as $s_t = \cos(t)$ for $t \in (-2\pi, 2\pi)$ interval, which drives an exponential transition function $G_t = 1 - \exp(-\lambda(s_t - c)^2)$ configured with a slope parameter $\lambda = 3$ and a threshold of $c = 0.3$. The structural design matrix X_t combined a periodic component $X_{t1} = \cos(\frac{1}{4}t)$ with a unitary intercept column $X_{t2} = 1$. By applying the specific parameter vectors $\beta_1 = (1, 2)$ and $\beta_2 = (2, 1)$, the dependent variable y_t was simulated as a non-linear combination of these states, incorporating an independent and identically distributed standard normal disturbance term e_t .

1.2 Code and results overview

```
t = linspace(-2*pi, 2*pi, 500);
s = cos(t);
lambd = 3;
c = 0.3;
g = 1-exp(-lambd*(s-c).^2);
X = [cos(t/4)', ones(length(t), 1)];
b1 = [1; 2];
b2 = [2; 1];
e = normrnd(0, 1, size(g'));
y = X*b1+(g'.*X)*b2+e;
figure;
plot(t, y, 'DisplayName', 'y(t)');
hold on;
plot(t, g, 'DisplayName', 'G(t)', 'LineWidth', 1.5);
title('Expo STR model');
legend;
param = [b1; b2; lambd; c];
L = lossExp(y, X, s, param);
disp('Loss L is below:')
L % 470.9648
```

Listing 1: Checking autocorrelation for 10 first lags

Quantitatively, the computed LS loss score of $L = 470.9648$ robustly validates the results of our model. The theoretical expectation for the SSE across a sample size of $n = 500$ observations converges strictly to 500. The empirical realization of approximately 471 falls comfortably within the acceptable variance of a Chi-squared distribution with 500 degrees of freedom.

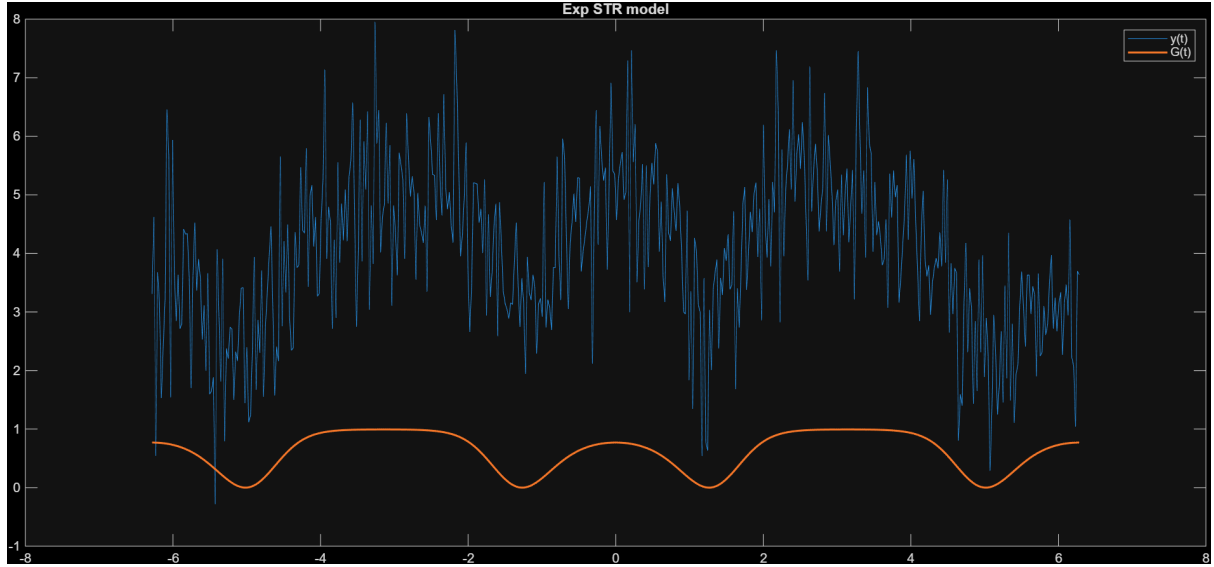


Figure 1: Simulated exponential STR model

Visually, the plot captures the complex, non-linear dynamics inherent to the specified STR model. The transition function G_t correctly exhibits the non monotonous dynamics of the data and symmetric oscillations characteristic of an exponential formulation driven by the periodic transition variable $s_t = \cos(t)$ over the defined domain $t \in (-2\pi, 2\pi)$. Moreover, the trajectory of the simulated dependent variable y_t demonstrates the appropriate regime-switching behavior. It smoothly changes between the base state defined by β_1 and the alternative state defined by the interaction with β_2 .

2 Task 4.2

2.1 Description

This task introduced the implementation of a comprehensive grid search algorithm to optimize the non-linear transition parameters of a STR model. The optimization procedure systematically evaluates a two dimensional parameter space, establishing a slope parameter grid of λ value comprising 50 logarithmically spaced values between 0.01 and 100, alongside a threshold parameter grid c consisting of 25 linearly spaced values bounded strictly by the 10th and 90th percentiles of the transition variable s_t . For every (λ, c) combination within this matrix, the structural model is formulated, thus enabling the OLS estimation of given vectors β_1 and β_2^* , from which the relative differential vector $\beta_2 = \beta_2^* - \beta_1$ is subsequently derived. The algorithm calculated and archived the LS loss score for each iteration to ultimately isolate the globally optimal (λ, c) combination that minimizes the objective loss.

2.2 Code and results overview

```

k = size(x, 2);
lambdas = logspace(-2, 2, 50);
cs = linspace(quantile(s, 0.1), quantile(s, 0.9), 25);
losses = nan(length(lambdas), length(cs));
for i = 1:length(lambdas) % grid search
    for j = 1:length(cs)
        lam = lambdas(i);
        c = cs(j);
        G = 1./(1+exp(-lam*(s-c)));
        Z2 = (1-G).*x;
        Z = G.*x;
        X2 = [Z2, Z];
        [betas_ols, ~] = OLS(X2, y);
        b1_ols = betas_ols(1:k);
        b2_star_ols = betas_ols(k+1:end);
        b2_ols = b2_star_ols - b1_ols;
        param_vector = [b1_ols', b2_ols', lam, c];
        losses(i, j) = loss(y, x, s, param_vector);
    end
end
[linear_idx, min_loss] = min(losses(:));
[opt_lam_idx, opt_c_idx] = ind2sub(size(losses), linear_idx);
lam_opt = lambdas(opt_lam_idx);
c_opt = cs(opt_c_idx);
G_opt = 1./(1+exp(-lam_opt*(s-c_opt)));
Z2_opt = (1-G_opt).*x;
Z_opt = G_opt.*x;
X2_opt = [Z2_opt, Z_opt];
[betas_opt, ~] = OLS(X2_opt, y);
b1_opt = betas_opt(1:k);
b2_star_opt = betas_opt(k+1:end);
b2_opt = b2_star_opt - b1_opt;

```

Listing 2: Checking autocorrelation for 10 first lags

Minimal loss score	1.2397e+03
Optimal lambda value	2.3300
Optimal c value	1.5104
Optimal b1 values	4.9322 10.0148 0.9904 19.9493
Optimal b2 value	30.2618 3.0299 5.0290 1.2512

The execution of the optimization procedure successfully isolated the global minimum of the objective function, yielding a minimized least squares loss score of $1.2397e + 03$. This optimal convergence occurs at a transition rate parameter of $\lambda = 2.3300$ and a structural threshold of $c = 1.5104$, which collectively characterize a moderate, smooth transition between the model's operational regimes rather than an abrupt structural break. Conditioned upon these globally optimal non-linear parameters, the corresponding estimation yielded a baseline regime vector of $\beta_1 = [4.9322, 10.0148, 0.9904, 19.9493]$ and a relative differential vector of $\beta_2 = [30.2618, 3.0299, 5.0290, 1.2512]$. The substantial magnitudes observed within the β_2 vector, particularly its primary coefficient, mathematically underscore that traversing the threshold c induces a highly significant structural adjustment to the underlying data-generating dynamics.

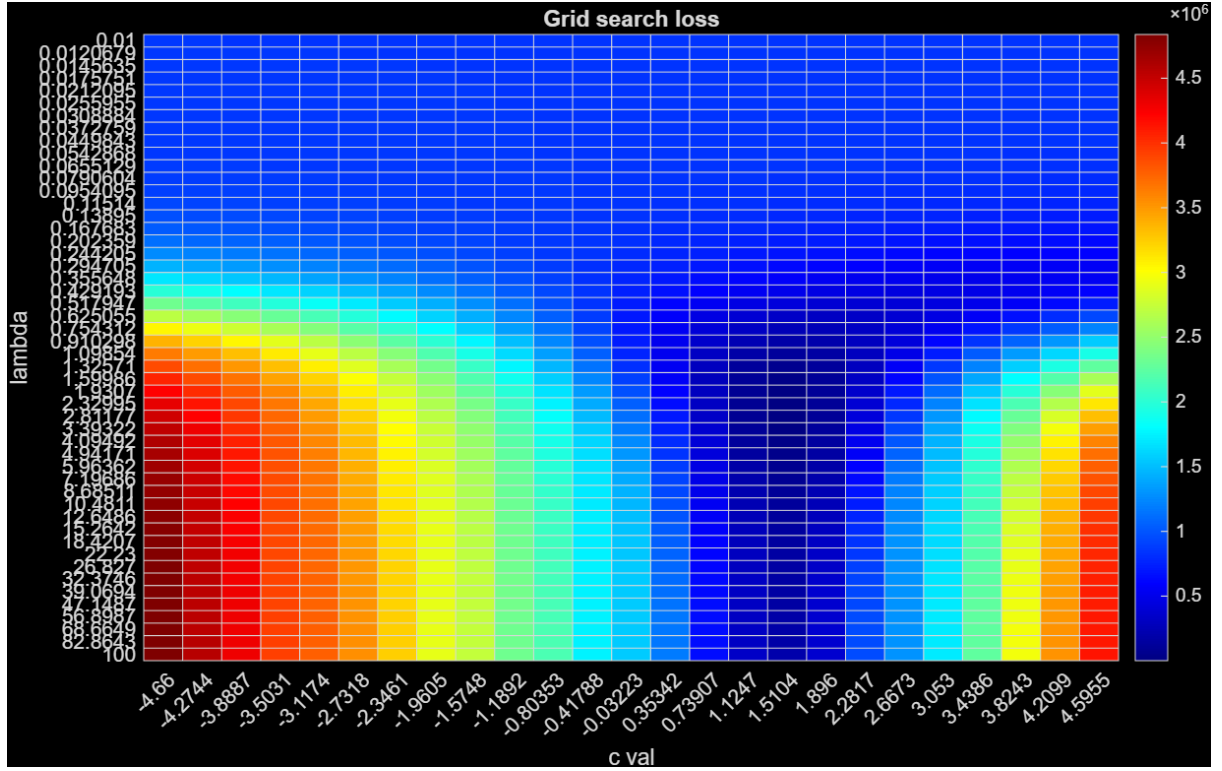


Figure 2: Task's 4.2 heatmap

This built heatmap provides a comprehensive visual diagnostic of the objective function's topology and the structural threshold. The loss surface is primarily characterized by a distinct convergence "valley" —easily recognized by its deep blue region. Notably, at heavily constrained, near-linear transition rates, the error metric remains uniformly stabilized across the entire domain of c , reflecting the mathematical muting of the structural break. However, as λ increases and the regime transition becomes more abrupt, the model's predictive accuracy becomes acutely sensitive to the threshold specification. Suboptimal threshold positioning, coupled with high transition rates, which are clearly visible in the intense red and yellow gradients occupying the lower left and lower right extremities, yields exponentially inflated error metrics approaching 4.5×10^6 . This severe parameter sensitivity validates the necessity of the 2D grid search approach, successfully navigating a highly non-linear optimization landscape to pinpoint the narrow corridor of structural stability.

3 Task 4.3

3.1 Description

This task focused on estimating the parameters of a STR model and evaluating different optimization configurations. Using the provided dataset the objective was to establish an initial parameter vector and iteratively refine it. First, the unconstrained optimization function "fminunc" was used to estimate the model's parameters. Following this, constrained optimization was applied using "fmincon", which restricts the transition rate parameter λ between 0.01 and 100, and bounds the threshold parameter c between the 10th and 90th percentiles of the transition variable. Finally, this constrained estimation was repeated with an adjusted Step Tolerance stopping criterion set to 0.05, culminating in a summary table that compares the resulting parameters and loss scores across all three estimation methods.

3.2 Code and results overview

```
load('Report4_3.mat');
params0 = [beta1; beta2; lam; c]';
fun = @(params) loss(y, x, s, params);
loss_initial = fun(params0);
disp('fminunc')
opts_unc = optimoptions('fminunc', 'MaxFunctionEvaluations', 5000, 'MaxIterations',
    2000);
optim_params_unc = fminunc(fun, params0, opts_unc);
loss_unc = fun(optim_params_unc);
disp('fmincon')
K = size(x, 2);
lb = [-inf(1, 2*K), 0.01, quantile(s, 0.1)];
up = [inf(1, 2*K), 100, quantile(s, 0.9)];
optim_params_con = fmincon(fun, params0, [], [], [], [], lb, up);
loss_con = fun(optim_params_con);
disp('fmincon & step tolerance of 0.05')
opts_con = optimoptions('fmincon', 'StepTolerance', 0.05);
optim_params_con_tol = fmincon(fun, params0, [], [], [], [], lb, up, [], opts_con);
loss_con_tol = fun(optim_params_con_tol);
% table with comparisons
Methods = {'Initial'; 'fminunc'; 'fmincon'; 'fmincon_StepTol'};
Params_Initial = params0;
Params_Unc = optim_params_unc;
Params_Con = optim_params_con;
Params_Con_Tol = optim_params_con_tol;
All_Params = [Params_Initial; Params_Unc; Params_Con; Params_Con_Tol];
Lambda_Est = [params0(end-1); optim_params_unc(end-1); optim_params_con(end-1);
    optim_params_con_tol(end-1)];
C_Est = [params0(end); optim_params_unc(end); optim_params_con(end);
    optim_params_con_tol(end)];
Loss_Scores = [loss_initial; loss_unc; loss_con; loss_con_tol];
ResultsTable = table(Methods, All_Params, Lambda_Est, C_Est, Loss_Scores);
disp('Comparison table')
disp(ResultsTable);
```

Listing 3: Checking autocorrelation for 10 first lags

Methods	All_Params								
Initial	-4.1401	2.232	19.352	-8.5426	8.209	62.38	1	9.2229	
fminunc	1.0299	2.0001	0.82076	10.295	4.9821	9.2152	2095.5	7.3634	
fmincon	1.0309	2.0001	0.8173	10.326	4.9806	9.1343	78.496	7.425	
fmincon_StepTol	1.0277	2.0002	0.82733	10.341	4.98	9.0929	73.266	7.4266	

Methods	Lambda_Est	C_Est	Loss_Scores
Initial	1	9.2229	1.0178e+06
fminunc	2095.5	7.3634	950.64
fmincon	78.496	7.425	950.03
fmincon_StepTol	73.266	7.4266	950.93

The optimization process demonstrates a significant improvement in model fit compared to the initial parameter guesses, which produced an extremely high loss score of 1.0178e+06. While the unconstrained algorithm fminunc drastically reduced the loss to 950.64, it yielded an invalid transition rate that vastly exceeds the prescribed upper bound of 100. By applying bounds, the fmincon method successfully corrected this, confining λ to 78.496 and the threshold c to 7.425 while achieving the lowest overall loss score of 950.03, proving it to be the most effective configuration for this model. Finally, introducing a Step Tolerance of 0.05 halted the algorithm slightly earlier in its search space, resulting in a marginal increase in the loss score and slightly shifted parameters $\lambda = 73.266$, $c = 7.4266$, clearly illustrating the inherent trade-off between computational stopping criteria and absolute mathematical precision.

4 Task 4.4

4.1 Description

The additional task expands the analysis to a real-world macroeconomic context by modeling GDP growth based on lagged unemployment, inflation, and renewable energy consumption. Using a versatile programming framework currently applied to a consolidated dataset for Poland, the procedure meticulously filters out historical missing values before estimating and comparing three distinct forecasting approaches: an STR model with a logistic transition function, an STR model with an exponential transition function and a baseline Linear Regression model. Both nonlinear STR frameworks utilize lagged unemployment as the transition variable and rely on constrained optimization fmincon to refine initial parameter estimates derived from a grid search. Ultimately, the performance of these three models is evaluated by plotting their predicted GDP trajectories alongside actual Polish GDP and quantitatively, utilizing the Bayesian Information Criterion to identify the model that provides the optimal balance of mathematical fit and simplicity.

4.2 Code and results overview

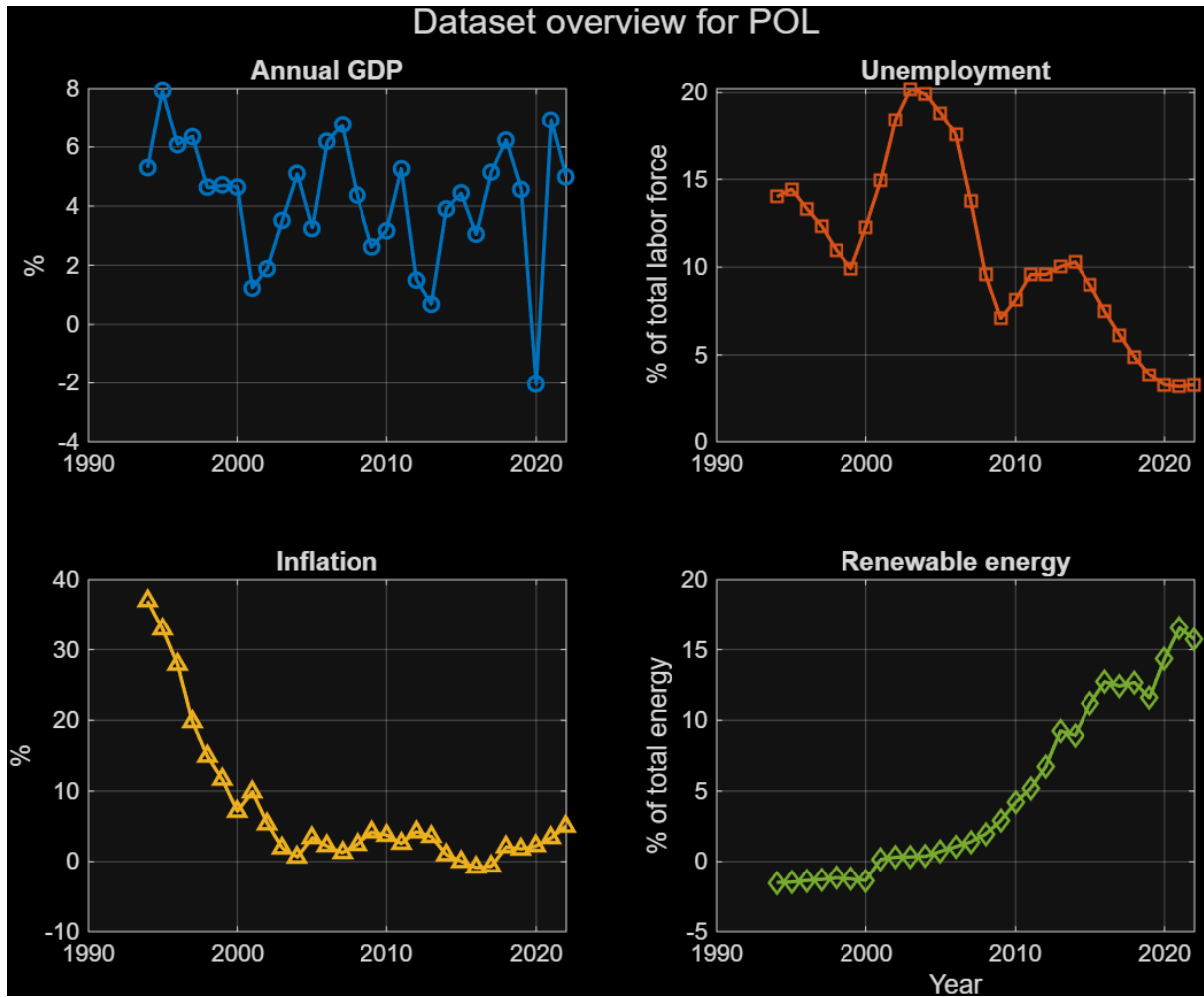


Figure 3: Task's 4.4 data overview

The macroeconomic overview for Poland from the 1990s to the 2020s visually captures a nation undergoing dramatic economic transformation. The early 1990s reflect a transition period, marked by high inflation that was aggressively brought under control and stabilized by the 2000s. Unemployment exhibits a cyclical but ultimately declining trajectory. After peaking in the early 2000s, the labor market steadily strengthened, pulling the jobless rate down significantly over the following two decades. Annual GDP growth remains highly dynamic throughout the entire period, characterized by cyclical peaks and sharp contractions that align clearly with major global economic shocks (such as coronavirus pandemic or 2008 financial crisis). Concurrently, the data reveals a stark structural shift in infrastructure, with renewable energy adoption remaining stagnant through the 1990s before initiating a steep, uninterrupted upward climb from the 2000s onward (related to European Union financing).

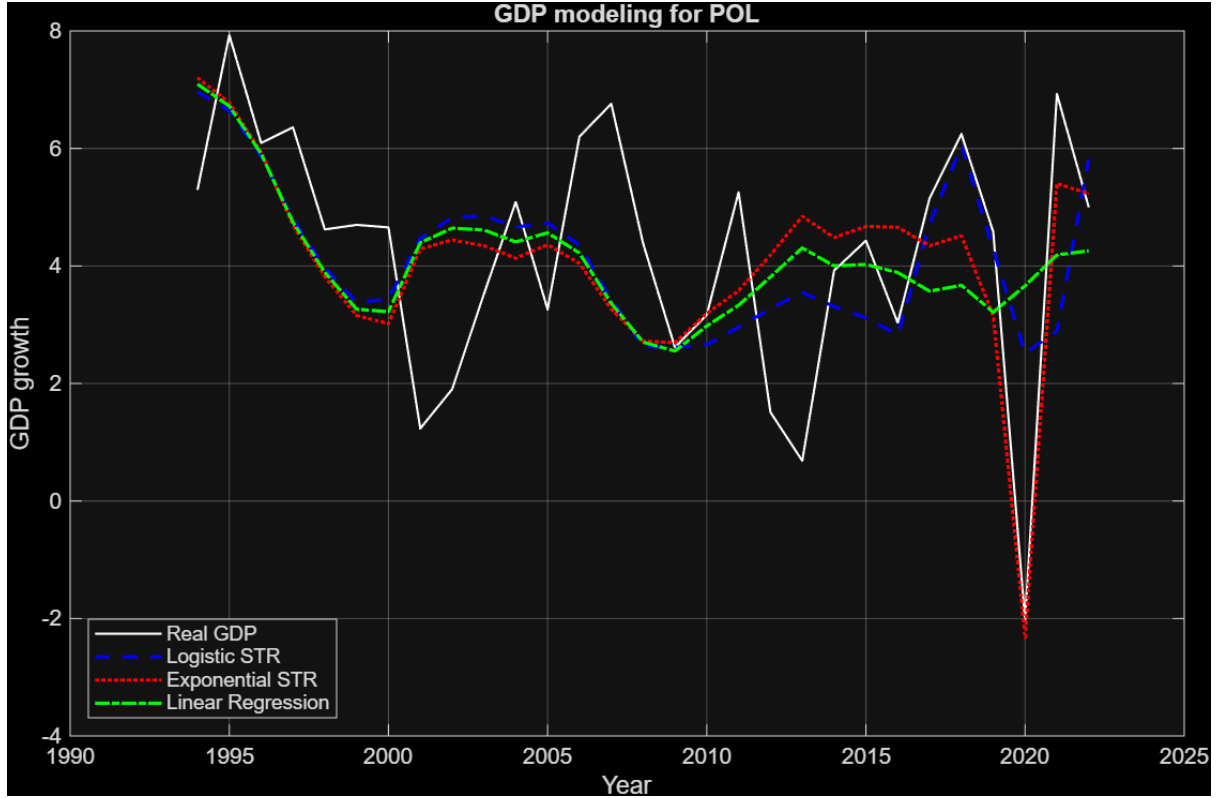


Figure 4: Chosen country's GDP modeling

The GDP modeling comparison successfully shows the limitations of linear approaches when forecasting volatile macroeconomic environments. The real GDP trajectory is highly erratic, and the baseline Linear Regression model acts largely as a smoothed average, completely failing to capture the magnitude of sudden economic peaks and troughs. In stark contrast, both nonlinear STR models demonstrate a much higher degree of flexibility. The Exponential STR model, in particular, stands out for its responsiveness to severe structural shocks, as it successfully mirrors the massive, abrupt contraction around 2020 that the linear model entirely ignores. This visual evidence clearly validates the use of transition variables to govern model parameters, proving that nonlinear frameworks are essential for accurately reflecting the realities of an economy susceptible to sudden and extreme shifts.

Models	RSS_Scores	BIC_Scores
Logistic STR	108.36	65.164
Exponential STR	90.184	59.84
Linear Regression	125.75	52.645

The final evaluation of this task in a form of comparison table presents a compelling narrative about the classic statistical trade-off between model fit and complexity. On one hand, the RSS heavily favors the nonlinear approaches. For example, the exponential STR model achieves the lowest error compared to the Linear Regression baseline, mathematically confirming its superior ability to track Poland's extreme economic fluctuations. However, the BIC – which strictly penalizes models for adding extra parameters – ultimately selects the Linear Regression model as the optimal choice with the lowest score of 52.645. This reveals a crucial insight, that while the STR models are undeniably better at capturing the complex, real-world volatility seen in the visual plots, the rigorous mathematical penalty of the BIC deems the simpler linear approach to be the most parsimonious, reminding us that visual fit to historical data does not always equate to the most statistically efficient predictive model.